KAIZEN MATHS

Lesson starter

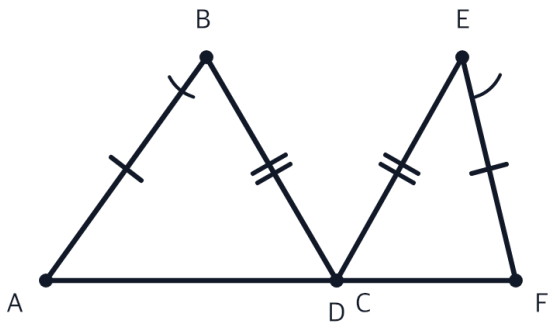
Formal Geometric Proof

Congruence & Similarity

Name _____

SA1

Prove the triangles congruent by matching side-angle-side in the correct order.



Use the markings to prove triangle ABC is congruent to triangle EFD.

GEO-EB04-L3 · SA

Show your working

KAIZEN MATHS

Lesson starter

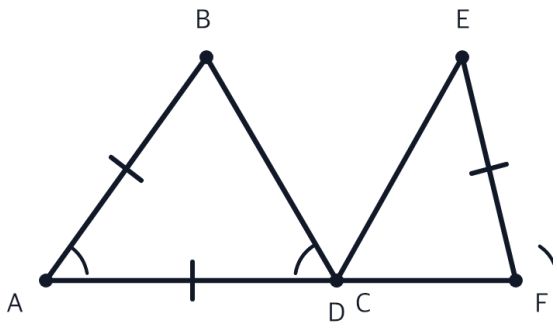
Formal Geometric Proof

Congruence & Similarity

Name _____

SB1

Prove the triangles congruent by matching angle-side-angle in the correct order.



Use the markings to prove triangle ABC is congruent to triangle DFE.

GEO-EB04-L3 · SB

Show your working

KAIZEN MATHS

Lesson starter

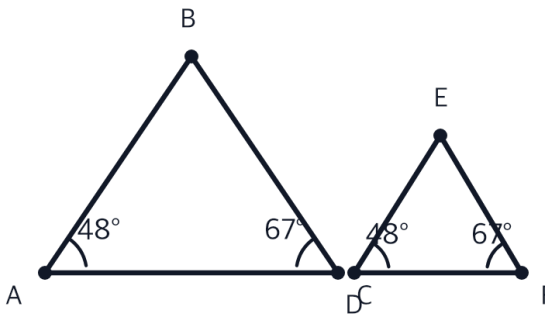
Formal Geometric Proof

Congruence & Similarity

Name _____

SC1

Prove the triangles similar by matching two equal angles.



Use the angle markings to prove triangle ABC is similar to triangle DEF.

GEO-EB04-L3 · SC

Show your working

KAIZEN MATHS

Lesson starter

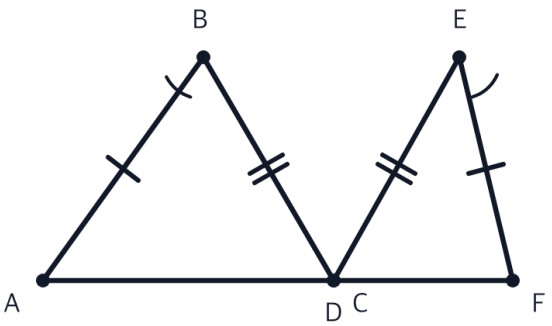
Formal Geometric Proof

Congruence & Similarity

Name _____

SD1

Prove congruence first, then use corresponding parts of congruent triangles.



The markings show triangle ABC is congruent to triangle EFD. Prove that AC = ED.

GEO-EB04-L3 · SD

Show your working

KAIZEN MATHS

Exit ticket

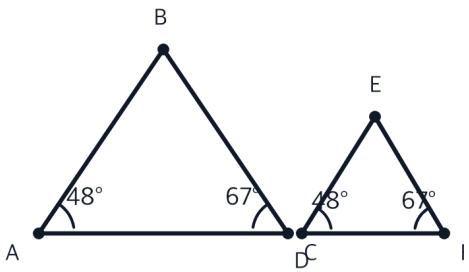
Formal Geometric Proof

Congruence & Similarity

Name _____

EA1

Prove the triangles similar by matching two equal angles.



Use the angle markings to prove triangle ABC is similar to triangle DEF.

Show your working

KAIZEN MATHS

Exit ticket

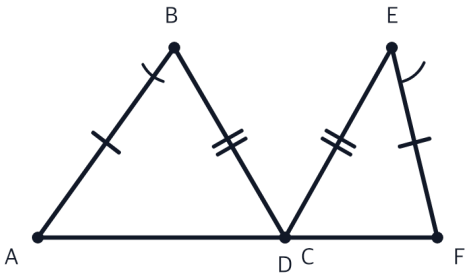
Formal Geometric Proof

Congruence & Similarity

Name _____

EB1

Prove congruence first, then use corresponding parts of congruent triangles.



The markings show triangle ABC is congruent to triangle EFD. Prove that AC = ED.

Show your working

KAIZEN MATHS

Exit ticket

Formal Geometric Proof

Congruence & Similarity

Name _____

EA1

Prove the triangles similar by matching two equal angles.

B
E
48°
67°
48°
67°
A
D
C
I

Use the angle markings to prove triangle ABC is similar to triangle DEF.

GEO-EB04-L3 · EA

Show your working

KAIZEN MATHS

Exit ticket

Formal Geometric Proof

Congruence & Similarity

Name _____

EB1

Prove congruence first, then use corresponding parts of congruent triangles.

B
E
A
D
C
F

The markings show triangle ABC is congruent to triangle EFD. Prove that AC = ED.

GEO-EB04-L3 · EB

Show your working

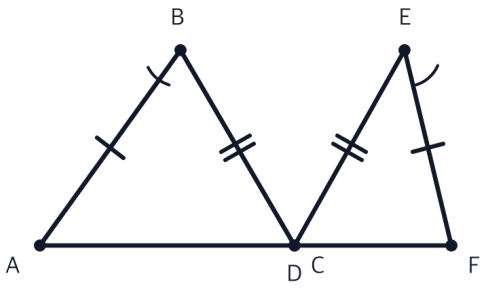
Teacher answers & working

Formal Geometric Proof · Congruence & Similarity

GEO-EB04-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1

Prove the triangles congruent by matching side-angle-side in the correct order.



Use the markings to prove triangle ABC is congruent to triangle EFD.

Answer:

Triangle ABC is congruent to triangle EFD by SAS.

Working:

Statement	Reason
$AB = EF$	Marked equal on the diagram.
$BC = FD$	Marked equal on the diagram.
Angle ABC = angle EFD	Marked equal on the diagram.
Triangle ABC is congruent to triangle EFD	Two sides and the included angle are equal, so the triangles are congruent by SAS.

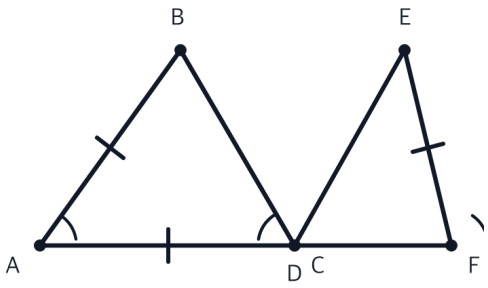
Teacher answers & working

Formal Geometric Proof · Congruence & Similarity

GEO-EB04-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1

Prove the triangles congruent by matching angle-side-angle in the correct order.



Use the markings to prove triangle ABC is congruent to triangle DFE.

Answer:

Triangle ABC is congruent to triangle DFE by ASA.

Working:

Statement	Reason
Angle BAC = angle FDE	Marked equal on the diagram.
$AC = DF$	Marked equal on the diagram.
Angle BCA = angle DFE	Marked equal on the diagram.
Triangle ABC is congruent to triangle DFE	Two angles and the included side are equal, so the triangles are congruent by ASA.

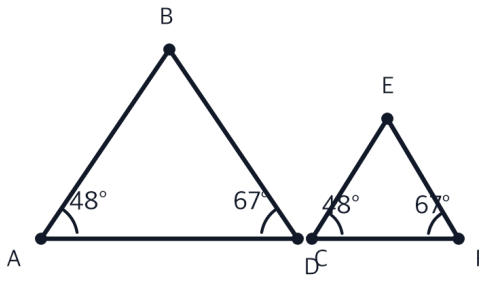
Teacher answers & working

Formal Geometric Proof · Congruence & Similarity

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SC1 / EA1

Prove the triangles similar by matching two equal angles.



Use the angle markings to prove triangle ABC is similar to triangle DEF.

Answer:

Triangle ABC is similar to triangle DEF by AA.

Working:

Statement	Reason
Angle BAC = angle EDF	Both angles are 48°.
Angle BCA = angle DFE	Both angles are 67°.
Triangle ABC is similar to triangle DEF	Two pairs of corresponding angles are equal, so the triangles are similar by AA.

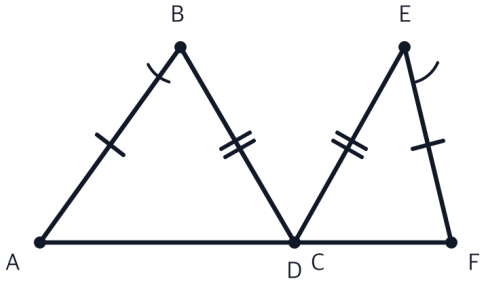
Teacher answers & working

Formal Geometric Proof · Congruence & Similarity

GEO-EB04-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1 / EB1

Prove congruence first, then use corresponding parts of congruent triangles.



The markings show triangle ABC is congruent to triangle EFD. Prove that AC = ED.

Answer:

$AC = ED$.

Working:

Statement	Reason
$AB = EF$	Marked equal on the diagram.
$BC = FD$	Marked equal on the diagram.
Angle ABC = angle EFD	Marked equal on the diagram.
Triangle ABC is congruent to triangle EFD	SAS congruence.
$AC = ED$	Corresponding parts of congruent triangles are equal.